

HAIYU ZHANG

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EDUCATION

Southern University of Science and Technology

2024 - 2026 (expected)

M.S. in Mathematics

GPA: 3.52/4.0 (89/100)

Courses: Advanced Probability, Convex Optimization Algorithms, Dynamical System, Machine Learning

Southern University of Science and Technology

2020 - 2024

B.S. in Mathematics

GPA: 3.68/4.0 (88/100)

Courses: Algebraic Topology, Differential Geometry, Abstract Algebra, Functional Analysis

University of California, Irvine

2023

Exchange Program

GPA: 4.0/4.0

Courses: Group Theory, Introduction to Programming Data Science, Geometry and Spacetime

PAPER

Topology-enhanced machine learning for consonant recognition

Preprint

Pingyao Feng*, Qingrui Qu*, **Haiyu Zhang**, Siheng Yi, Zhiwang Yu, Zeyang Ding, Yifei Zhu

RESEARCH EXPERIENCE

Fractal Structure and Generalization Bounds with Intrinsic Dimension for Stochastic Optimization Algorithms

2025 - Present

Master Thesis, Advisor: Yifei Zhu

- Derive a generalization bound based on the intrinsic hypothesis complexity and algorithm stability.
- Explicitly characterize how these two quantities depend on the hyperparameters of the stochastic optimization algorithm to better understand its implicit regularization effects.
- Explain the phenomenon of grokking by tracking the evolution of intrinsic dimension of weight trajectories and data representations.

Topology-enhanced machine learning for consonant recognition

2024 - 2025

Research Project, Key Contributor

- Proposed a topological methodology for time series analysis using time-delay embedding and persistent homology to extract topological acoustic features from speech signals.
- Enhanced the recurrent neural networks with topological features and applied the new framework to consonant classification. Compared it with state-of-the-art models and achieved superior performance in terms of accuracy, steadiness, and robustness to noise.

Topological Data Analysis and Topological Deep Learning with Applications to Image Data 2023 - 2024

Undergraduate Thesis, Advisor: Yifei Zhu

- Reproduced the work of Gabrielsson and Carlsson on topology of neural networks by applying persistent homology to study the topological manifolds of Prewitt and Laplacian convolution kernels in image classification model.
- Proposed an optimized topological CNN model for image classification based on the work of Love et al., using convolution kernels sampled from distributions on these manifolds.

PROJECT

Deep Parametric Portfolio Policies with Nonlinear Feature Learning2025

Group Course Project, Key Contributor

Developed a framework combining autoencoder-based nonlinear feature learning with Parametric Portfolio Policies (PPP) to address the challenges of high-dimensional stock features and limitations of traditional linear methods in prior portfolio constructions.

AWARDS

Outstanding Teaching Assistant Award2024

Outstanding Undergraduate Thesis Award2024

Honorable Mention of Mathematical Contest in Modeling (MCM) in the United States2023

Third Class of the Merit Student Scholarship2021

Second Class of the Freshmen Scholarship2020

SEMINARS AND PRESENTATIONS

Applied and Computational Topology SeminarFall 2023 - Present

Included topics in topological data analysis and topological deep learning such as persistent homology, along with their applications in data science.

Dynamical System SeminarSpring 2024

Included the theory of iteration of expanding maps and topics in geometric measure theory of the underlying invariant fractal sets.

CONFERENCES

International Workshop on Algebraic Topology (IWoaT)2025

The 4th Annual Centre for Topological Data Analysis Conference in Oxford (Spires 2024)2024

Greater Bay Area Topology Conference2024

TEACHING

Teaching assistant for Calculus2024

Teaching assistant for Applied and Computational Topology2025

SKILLS

Code: Python (Practised) Matlab (Practised) Java (Beginner) Julia (Beginner)

Edit: LaTeX (Practised)

Languages: English (IELTS 7.0) Mandarin (Native)

PERSONAL INTERESTS

Psychology Classical music Badminton and table tennis